Aleksandr Medvedev

LLMs | AI Agents | RAG | MLOps

Moscow | 30+ h/week | medv.ax@outlook.com | github.com/MedvAx-AI | medvax-ai.github.io

SUMMARY

3rd-year Applied Artificial Intelligence student building production-oriented AI systems across LLM agents, RAG pipelines, APIs, data/ML workflows, deployment, and continuous evaluation. Built an auditable LLM routing service evaluated across 1,000 Qwen2.5 runs on 500 MMLU questions, reducing p50 latency by 15%. Automated an end-to-end retraining and deployment pipeline every 5 minutes with 10 CI checks.

SKILLS

AI Engineering: Python, FastAPI, REST APIs, JSON, HTTP, SQL/PostgreSQL, Git, GitHub Actions, Docker

LLM & Agents: LLMs, AI agents, RAG pipelines, Hugging Face Transformers, FAISS, human-in-the-loop systems

ML & Data: PyTorch, TensorFlow, Scikit-learn, Pandas, NLP, text embeddings, semantic search

MLOps: DVC, MLflow, automated retraining/deployment, CI validation, reproducible model releases

PROJECTS

BudgetRoute-LLM - LLM Routing Agent with RAG & Human Review

Python | PyTorch | FastAPI | FAISS | PostgreSQL | github.com/MedvAx-AI/budgetroute-llm

- **Reduced p50 inference latency by 15% (390 ms vs 461 ms)** by routing 78% of requests to Qwen2.5-0.5B while selecting between small/large models, RAG retrieval, cascade, abstention, and human review.
- **Built an auditable AI service** by combining FastAPI, PostgreSQL, traceable RAG evidence (chunk ID, score, source), a human-review queue, tamper-evident audit chain, and control plane.
- **Achieved 1,000/1,000 reproducible answers on a clean rerun** by using leakage-free 300/100/100 splits, pinned model revisions, and bootstrap confidence intervals for held-out evaluation.

Yacht Resistance MLOps Pipeline

Scikit-learn | Pandas | DVC | MLflow | Docker | FastAPI | github.com/MedvAx-AI/pmldl-yacht-mlops

- **Eliminated manual retraining and deployment steps** by automating DVC -> MLflow -> FastAPI/Streamlit every 5 minutes, verified by 10 CI tests and model SHA-256 plus MLflow run-ID checks.
- **Achieved R2 0.993 and RMSE 1.28 on 70 observations from 5 unseen hulls** by training an Extra Trees pipeline with hull-grouped splits and automatic baseline validation.

Beyond Proximity / SemanticSplat - Research Team Lead

- **Led a 4-person team to complete a 150-query evaluation across 5 scenes** by coordinating experimental execution plus ScanNet and Replica pilot studies.

EDUCATION

Innopolis University | BSc Applied Artificial Intelligence | 2024-2028